

INDEPENDENT CONSULTANT'S
SAFETY INSPECTION REPORT

ELK BASIN DAM

FERC Project No. 4182

Elk Basin Irrigation Project
Elk Creek, Elmore County, Idaho

Part 12D — Five-Year Periodic Inspection
Inspection Date: September 14-15, 2020

Prepared for:
Elk Basin Irrigation District

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Report Date: October 30, 2020

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1.0 INTRODUCTION

This report presents the findings of the independent consultant's safety inspection of Elk Basin Dam, performed in accordance with the requirements of 18 CFR Part 12, Subpart D. The inspection was conducted on September 14 and 15, 2020, by Cascade Dam Safety Consultants, LLC (CDSC) on behalf of the Elk Basin Irrigation District (EBID), the FERC licensee. This is the eleventh five-year periodic inspection since the FERC Part 12D program was established for this project in 1970.

The inspection included a visual examination of all dam components, review of instrumentation and monitoring data, review of the Emergency Action Plan (EAP), and assessment of the licensee's dam safety program. The previous Part 12D inspection was conducted in September 2015 by the same firm; reference is made to that report for baseline conditions. This report focuses on changes in condition since the 2015 inspection and identifies items requiring attention.

1.1 Documents Reviewed

The following documents were reviewed in preparation for and during the inspection:

- USBR As-Built Construction Report, Elk Basin Dam (December 1968)
- USBR Design Data Report, Elk Basin Dam (1965)
- Previous FERC Part 12D Inspection Reports (2010, 2015)
- EBID Annual Dam Safety Inspection Reports (2016-2020)
- Instrumentation monitoring data (2015-2020)
- Surveillance and Monitoring Plan (revised 2018)
- Emergency Action Plan (revised 2019)
- FERC Engineering Guidelines for Dam Safety, Chapter 11 (Arch Dams)
- USSD Guidelines on Concrete Dams Affected by ASR (2019)

2.0 PROJECT DESCRIPTION

Elk Basin Dam is a 145-foot-high concrete arch dam with earthen embankment abutment transitions, impounding Elk Creek in the Salmon River Mountains of central Idaho. The dam was constructed by the U.S. Bureau of Reclamation between 1966 and 1968 and provides 18,500 acre-feet of storage for the Elk Basin Irrigation Project. A 2.4-megawatt hydroelectric powerhouse operates at the dam toe under FERC Project No. 4182. The dam is 52 years old at the time of this inspection. A full description of the dam and its appurtenant structures is provided in the previous inspection reports and the USBR As-Built Construction Report (1968); only a brief summary is provided here.

3.0 INSPECTION FINDINGS

3.1 Concrete Arch — Alkali-Silica Reaction

The downstream face of the concrete arch exhibits widespread map cracking and gel exudation consistent with alkali-silica reaction (ASR). The cracking was first documented during the 2010 Part 12D inspection as isolated hairline cracks on the downstream face between elevations 6,180 and 6,210 (the lower third of the arch). By the 2015 inspection, the cracking had progressed both in extent and severity, with gel deposits (white to translucent) observed at several crack intersections.

At the current 2020 inspection, the following conditions were observed:

- Map cracking is now present over approximately 60 percent of the downstream face area, extending from the base to approximately elevation 6,220 (lower two-thirds of the arch). Crack widths range from hairline to 0.04 inches, with the widest cracks in the lower third of the arch.
- Gel exudation is visible at approximately 25 locations, predominantly at crack intersections and along construction lift joints. The gel is white to amber in color and appears fresh (not desiccated), indicating active ASR.
- The upstream face shows significantly less cracking, limited to isolated hairline cracks at construction joints. This asymmetry is typical of ASR-affected dams, as the upstream face is saturated and cooler, while the downstream face experiences wetting-drying and thermal cycles that accelerate ASR gel expansion.
- No structural distress (block displacement, arch deformation, or joint opening beyond normal thermal movement) was observed. The arch continues to function as designed from a structural perspective.

The ASR is attributable to the use of locally sourced rhyolite gravel as coarse aggregate in the original concrete mix. The 1964 petrographic examination identified the aggregate as "potentially reactive," but the significance of this finding was not fully appreciated given the state of practice at the time of construction (1966-1968). The cement alkali content (0.72 percent Na₂O equivalent) exceeds the current ASTM C150 limit of 0.60 percent for use with reactive aggregates.

Assessment: The ASR is a long-term degradation mechanism that reduces the tensile strength and elastic modulus of the affected concrete. While the arch currently retains adequate structural capacity, the progression of ASR over the past decade indicates that the condition will continue to worsen. Core sampling and testing are recommended to quantify the depth of ASR penetration, residual concrete strength, and remaining service life. This information is essential for evaluating remedial alternatives.

3.2 Left Abutment — Increasing Seepage

Seepage measured at the left abutment toe drain weir has increased from approximately 12 gallons per minute (gpm) in 2012 to 28 gpm in September 2020, representing a 133 percent increase over 8 years. The seepage was 8 gpm at first filling (1969) and remained relatively stable between 8 and 14 gpm from 1969 through 2012. The recent increase warrants investigation.

Table 3-1: Historical Seepage Summary (at or near full pool)

Year	Left Abt. Seepage (gpm)	Right Abt. Seepage (gpm)	Reservoir Elev. (ft)	Notes
1969	8	5	6,240	First filling, baseline
1975	10	5	6,238	Normal operation
1985	11	6	6,239	Stable
1995	12	6	6,240	Slight increase noted
2005	12	6	6,238	Stable
2010	13	7	6,239	Slight increase, 12D noted
2012	12	6	6,237	Last stable reading
2015	18	7	6,240	Increasing trend begins
2018	23	7	6,239	Continuing increase
2020	28	7	6,240	Current reading

Key observations regarding the seepage increase:

- The increase is specific to the left abutment; right abutment seepage has remained stable at 5 to 7 gpm since 1969.
- The seepage increase does not correlate with reservoir level changes (reservoir has operated at similar levels throughout the period).
- Piezometer P-3 (left abutment foundation, elevation 6,185) shows a gradual increase in pore pressure from 8.2 psi in 2012 to 14.5 psi in 2020, consistent with the seepage increase.
- The seepage water is clear with no visible turbidity, fines, or discoloration. This is a positive indicator that internal erosion (piping) of the embankment core is not currently occurring.
- The gradual nature of the increase (over 8 years) suggests a slow deterioration of the grout curtain or opening of foundation discontinuities, rather than a sudden failure mechanism.

Assessment: The seepage increase is a Potential Failure Mode (PFM) indicator and warrants investigation to determine the seepage path, assess the condition of the embankment core and foundation contact, and evaluate whether the seepage can be controlled by supplemental grouting. A boring through the left abutment embankment into the foundation is recommended.

3.3 Right Abutment — Crest Displacement

Survey monument data indicate that the right abutment embankment crest has moved approximately 3.0 inches in the downstream direction over the 15-year period from 2005 to 2020. The displacement was first identified during the 2010 Part 12D inspection, at which time approximately 1.5 inches of cumulative downstream displacement had been measured since the survey baseline was established in 1995. The displacement rate appears to be approximately constant at 0.2 inches per year.

Table 3-2: Right Abutment Crest Displacement (Downstream, Cumulative)

Year	Monument RA-1 (in)	Monument RA-2 (in)	Monument RA-3 (in)	Monument RA-4 (in)	Avg. (in)
1995	0.00	0.00	0.00	0.00	0.00
2000	0.42	0.38	0.51	0.35	0.42
2005	0.95	0.88	1.12	0.82	0.94
2010	1.48	1.35	1.72	1.25	1.45
2015	2.02	1.88	2.35	1.78	2.01
2020	2.58	2.42	3.02	2.30	2.58

Key observations:

- The displacement is concentrated in the central portion of the embankment (monuments RA-2 and RA-3, near the maximum section). Monument RA-3 shows the largest displacement (3.02 inches), consistent with its location at the highest section.
- The displacement rate is approximately linear (constant) at 0.2 inches per year. There is no evidence of acceleration, which would indicate progressive failure.
- Vertical settlement of the crest is less than 0.5 inches over the same period, which is not unusual for a 75-foot-high embankment after 50+ years.
- Embankment piezometers in the right abutment show stable pore pressures with no increasing trend, suggesting the displacement is not driven by increasing pore pressures.
- No cracking, bulging, or seepage anomalies are visible on the downstream face of the right abutment embankment.

Assessment: The constant-rate downstream displacement is most likely attributable to long-term creep in the glacial till foundation and/or the embankment core material under the sustained load of the embankment. While the current rate and magnitude do not indicate imminent instability, the total displacement (3 inches) exceeds the design predictions, and the cause should be confirmed through subsurface investigation. A boring through the right abutment embankment into the foundation is recommended to characterize foundation conditions and provide data for an updated slope stability analysis.

3.4 Other Components

Spillway: The uncontrolled ogee spillway on the right abutment is in satisfactory condition. No cracking, erosion, or structural distress was observed. The stilling basin shows minor abrasion from infrequent spills but remains functional. No immediate action required.

Outlet Works: The 48-inch steel outlet conduit and control valves were inspected and operated during the inspection. All gates are functional. Minor surface corrosion was noted on the downstream valve body; recoating is recommended within 2 years. The conduit interior was not inspected (dewatered inspection recommended at next maintenance shutdown).

Powerhouse: The powerhouse structure and penstock are in good condition. Turbine and generator maintenance is current. No geotechnical concerns.

Instrumentation: Of the original instrumentation installed in 1968, approximately 65 percent remains functional. Non-functional instruments include 8 of 24 concrete strain meters (failed due to age), 4 of 18 thermometers, and 2 of 12 foundation piezometers. The remaining instruments continue to provide useful data. The plumb lines, joint meters, survey monuments, embankment piezometers, and seepage weirs are all functional.

4.0 CONDITION RATINGS AND RISK CLASSIFICATION

Table 4-1: Component Condition Ratings

Component	2015 Rating	2020 Rating	Trend	Recommended Action
Concrete arch — structural	Satisfactory	Satisfactory	Stable	Monitor
Concrete arch — ASR	Fair	Poor	Worsening	Core sampling, strength testing
Left abutment embankment	Satisfactory	Fair	Declining	Investigate seepage path
Left abutment seepage	Fair	Poor	Worsening	Boring, piezometer, grouting eval.
Right abutment embankment	Fair	Fair	Stable	Investigate displacement cause
Right abutment displacement	Fair	Fair	Stable rate	Boring, stability analysis update
Spillway	Satisfactory	Satisfactory	Stable	Routine maintenance
Outlet works	Satisfactory	Satisfactory	Stable	Recoat valve, inspect conduit
Instrumentation	Fair	Fair	Declining	Replace failed instruments
Foundation (arch)	Satisfactory	Satisfactory	Stable	None
Emergency Action Plan	Satisfactory	Satisfactory	N/A	Update per findings

Rating scale per FERC Engineering Guidelines: Satisfactory = meets design intent, no significant deficiencies; Fair = minor to moderate deficiencies, investigation or monitoring needed; Poor = significant deficiencies, corrective action required; Unsatisfactory = serious deficiencies, immediate action required.

Overall Dam Condition Rating: FAIR — The dam is safe for continued operation under normal loading conditions. However, three components have declining or poor ratings that require investigation within the next Periodic Facility Review cycle. The overall hazard potential classification remains **HIGH** based on downstream population at risk and economic consequences of failure.

5.0 RECOMMENDATIONS

5.1 Priority 1 — Periodic Facility Review Investigation (2024-2025)

CDSC recommends that the licensee initiate a comprehensive Periodic Facility Review (PFR) investigation to address the three primary concerns identified in this inspection. The investigation should include:

a) Concrete Arch Core Drilling: One vertical core boring through the arch dam section near the crown (maximum arch stress zone). The boring should extend from the dam crest through the full arch thickness and into the foundation rock. Core specimens should be tested for compressive strength (ASTM C42), split tensile strength (ASTM C496), and petrographic analysis (ASTM C856) to quantify ASR depth, severity, and residual strength at multiple depths. The concrete/rock foundation contact should be inspected for bond condition and deterioration.

b) Left Abutment Boring: One boring through the left abutment embankment from the crest to a minimum depth of 50 feet into the foundation bedrock. The boring should penetrate all embankment zones (shell, core, shell), the foundation contact, the weathered bedrock zone, and into fresh bedrock. Samples should be obtained for permeability testing, classification, and strength testing. A vibrating wire piezometer should be installed in the foundation bedrock to characterize pore pressures along the suspected seepage path.

c) Right Abutment Boring: One boring through the right abutment embankment to a minimum depth of 25 feet into the foundation bedrock. The boring should characterize the glacial till foundation, the till/bedrock contact, and the bedrock below the embankment. Samples should be tested for strength (triaxial CU), classification, and consolidation/creep properties. An inclinometer casing should be installed to monitor the depth and rate of displacement.

5.2 Priority 2 — Monitoring Enhancements (2021-2022)

- Increase left abutment seepage measurement frequency from monthly to weekly until the PFR investigation is completed.
- Increase right abutment survey monument readings from semi-annual to quarterly.
- Install a turbidity sensor at the left abutment seepage weir to provide early warning of internal erosion.
- Replace failed foundation piezometers P-8 and P-11 (arch section) to restore uplift monitoring coverage.

5.3 Priority 3 — Routine Maintenance (Ongoing)

- Recoat downstream outlet valve body within 2 years.
- Perform dewatered inspection of the outlet conduit interior during next scheduled maintenance shutdown.
- Update the Emergency Action Plan to reflect current downstream development and contact information.

6.0 CONCLUSIONS

Elk Basin Dam is rated in FAIR overall condition. The dam is safe for continued operation under normal and flood loading conditions. However, three issues — progressive ASR in the arch concrete, increasing seepage at the left abutment, and ongoing displacement of the right abutment embankment — require investigation to assess long-term safety and develop remedial strategies.

The recommended PFR investigation (Section 5.1) should be initiated within 3 to 4 years (by 2024-2025) to characterize these conditions and inform remedial design. In the interim, enhanced monitoring (Section 5.2) will provide early warning of any change in condition. None of the identified issues require emergency action or restriction of reservoir operations at this time.

The licensee's dam safety program, including the Surveillance and Monitoring Plan, Emergency Action Plan, and operations and maintenance practices, is adequate and consistent with FERC requirements. The licensee is commended for its diligent monitoring and maintenance of the facility over its 52-year history.

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